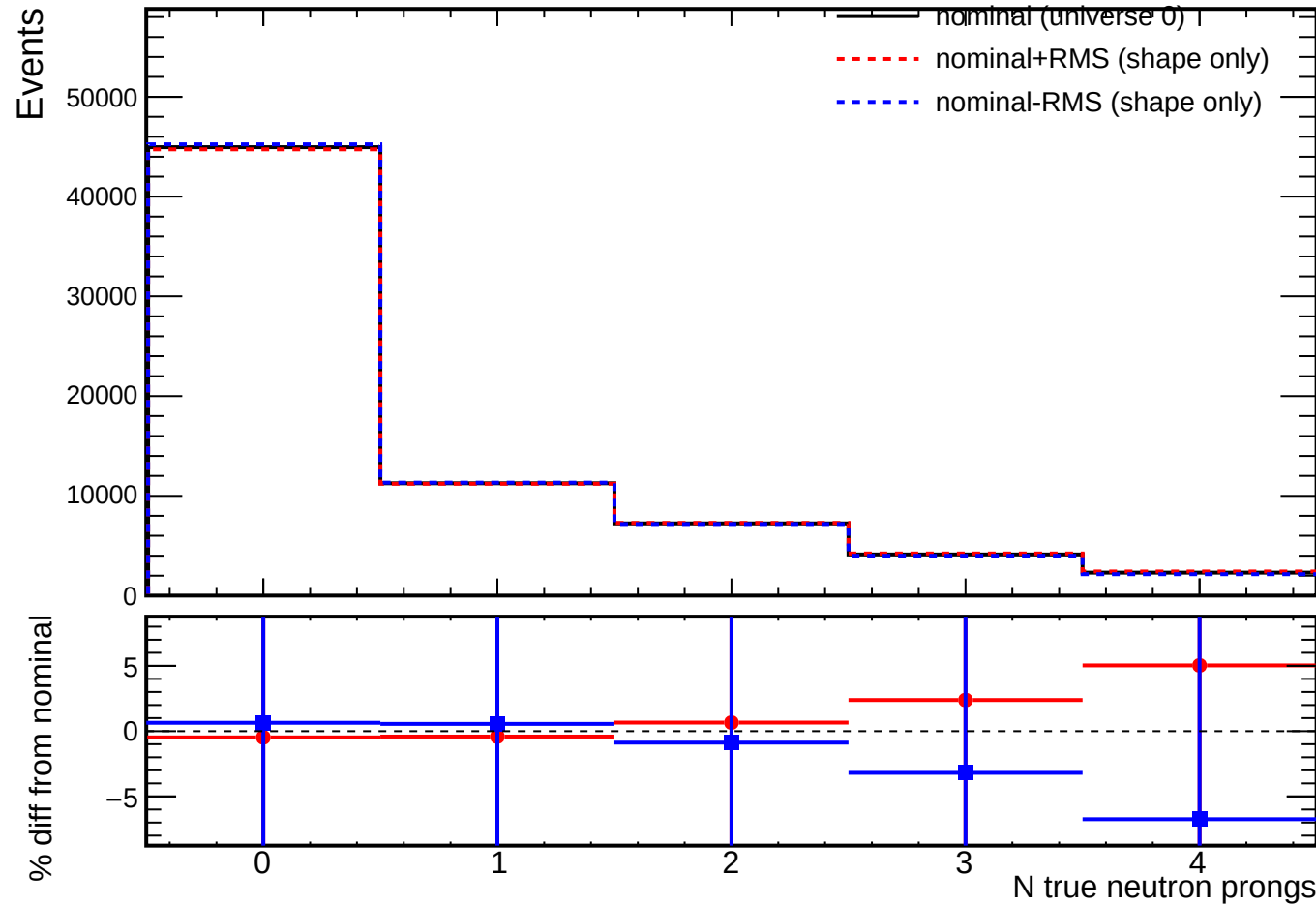
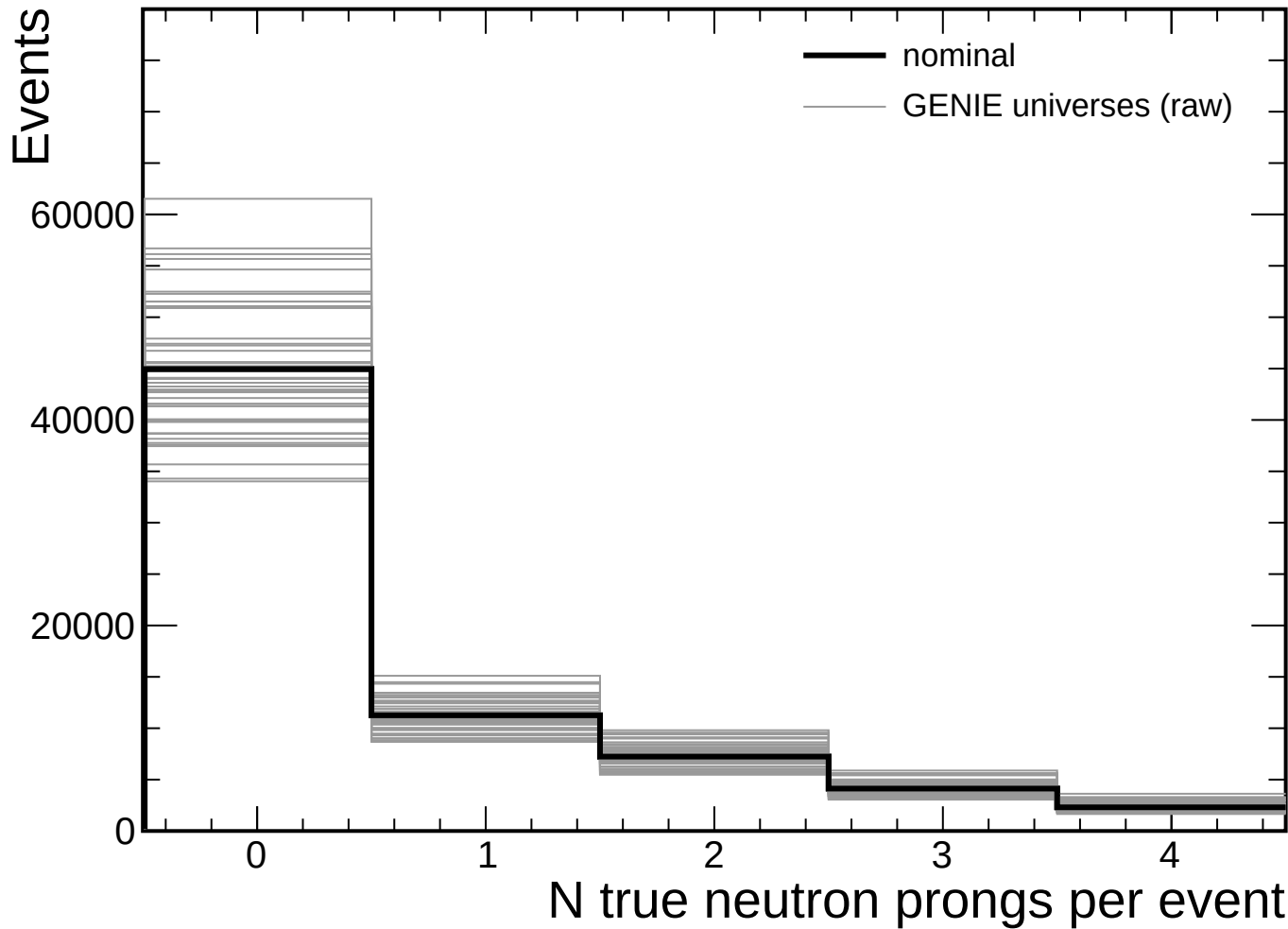
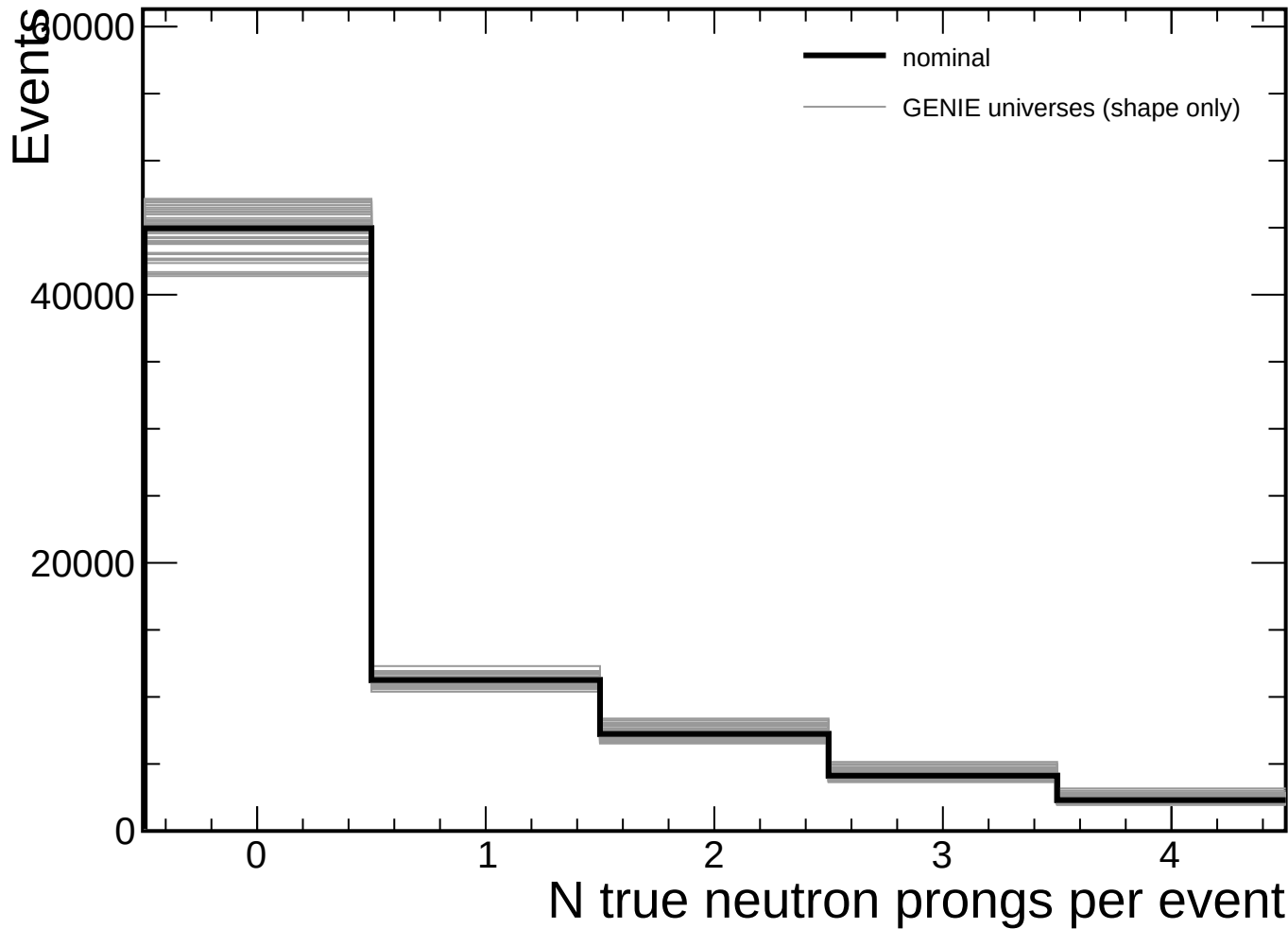


FHC: N true neutron prongs -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=9.72$ $\chi^2_{\text{shape},-\text{RMS}}=17.44$ $\Delta\langle x \rangle_{+\text{RMS}}=+0.0116$ $\Delta\langle x \rangle_{-\text{RMS}}=-0.0155$
 $\chi^2_{\text{raw},+\text{RMS}}=1483.33$ $\chi^2_{\text{raw},-\text{RMS}}=1483.33$ rate: +14.51% / -14.51%



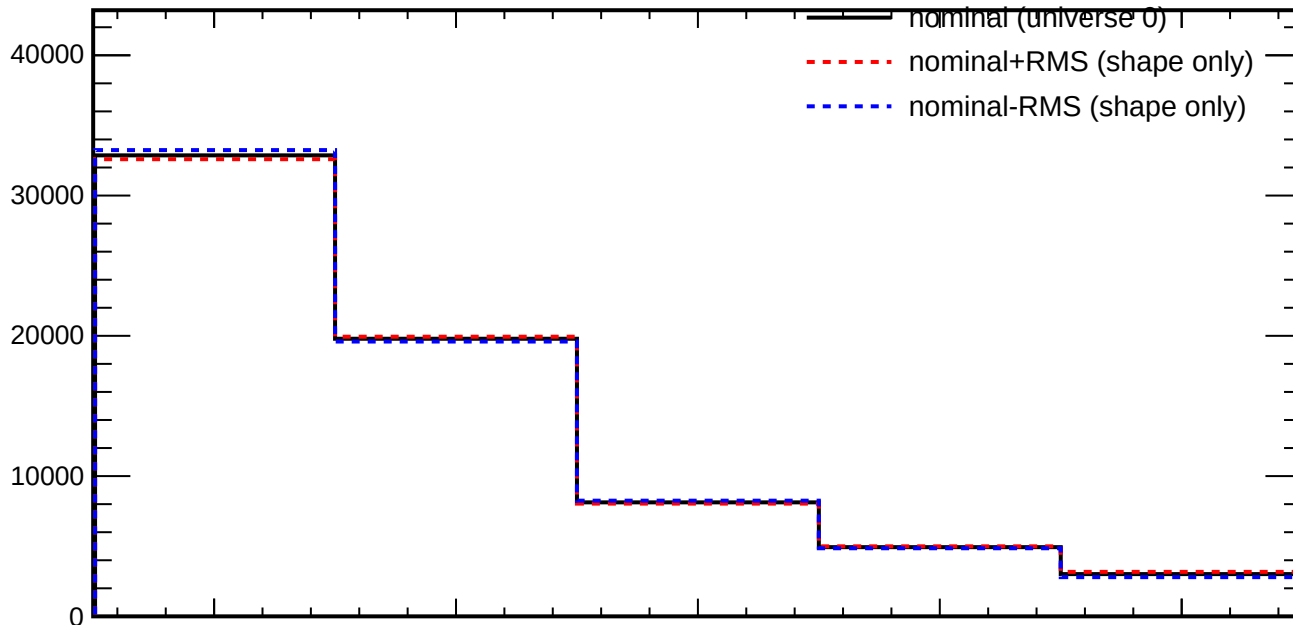
FHC: N true neutron prongs GENIE multiverse (26 relevant systs, raw, 50 universes)



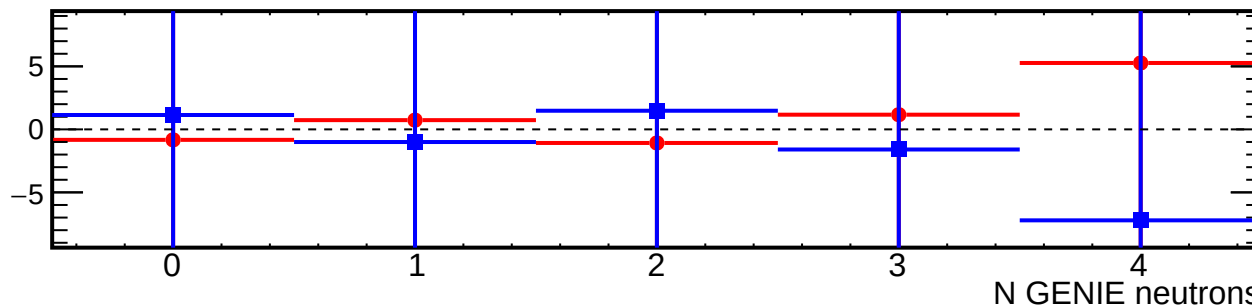


FHC: N GENIE neutrons -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=13.32$ $\chi^2_{\text{shape},-\text{RMS}}=25.01$ $\Delta\langle x \rangle_{+\text{RMS}}=+0.0113$ $\Delta\langle x \rangle_{-\text{RMS}}=-0.0155$
 $\chi^2_{\text{raw},+\text{RMS}}=1694.57$ $\chi^2_{\text{raw},-\text{RMS}}=1694.57$ rate: +15.62% / -15.62%

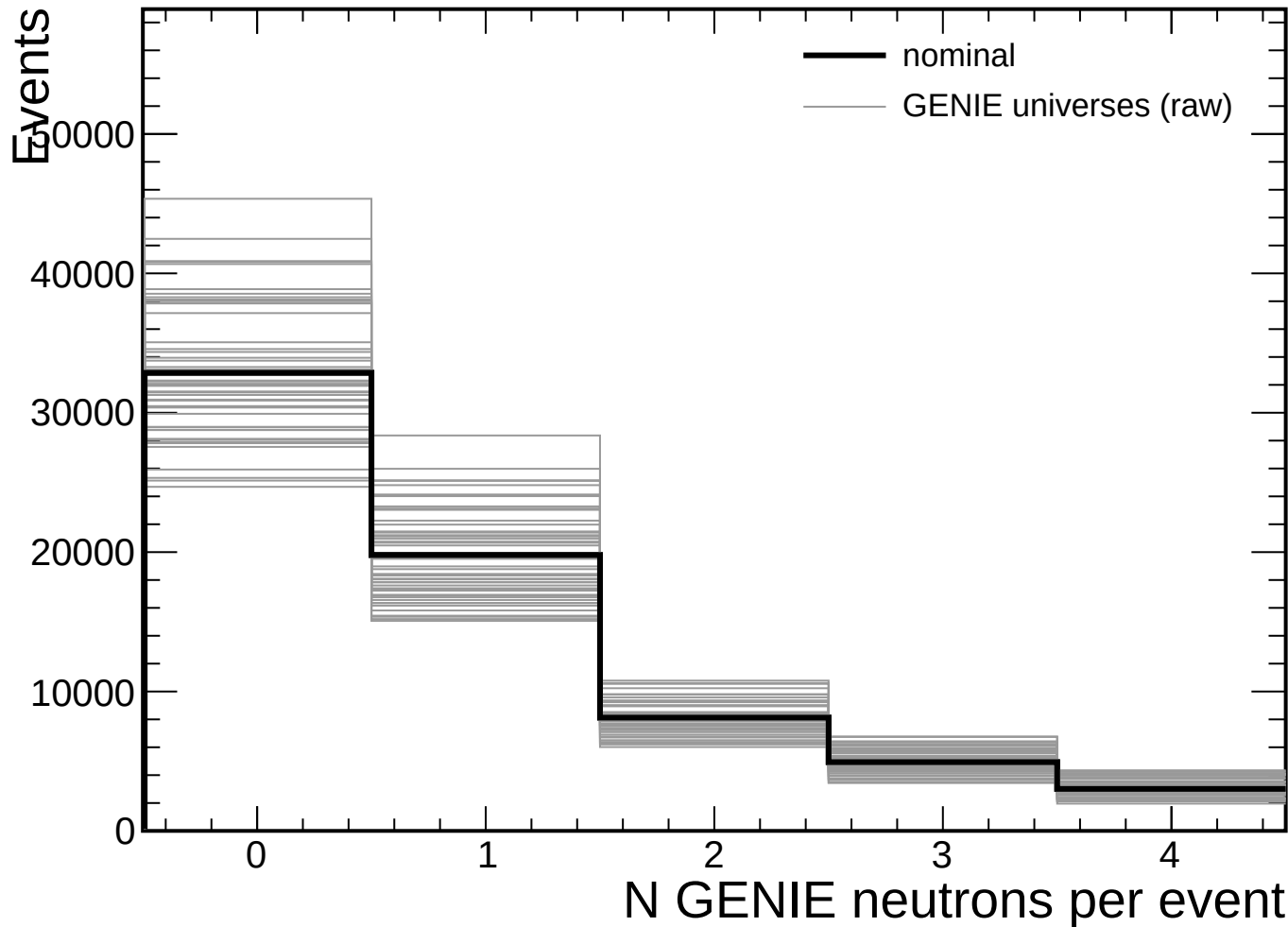
Events

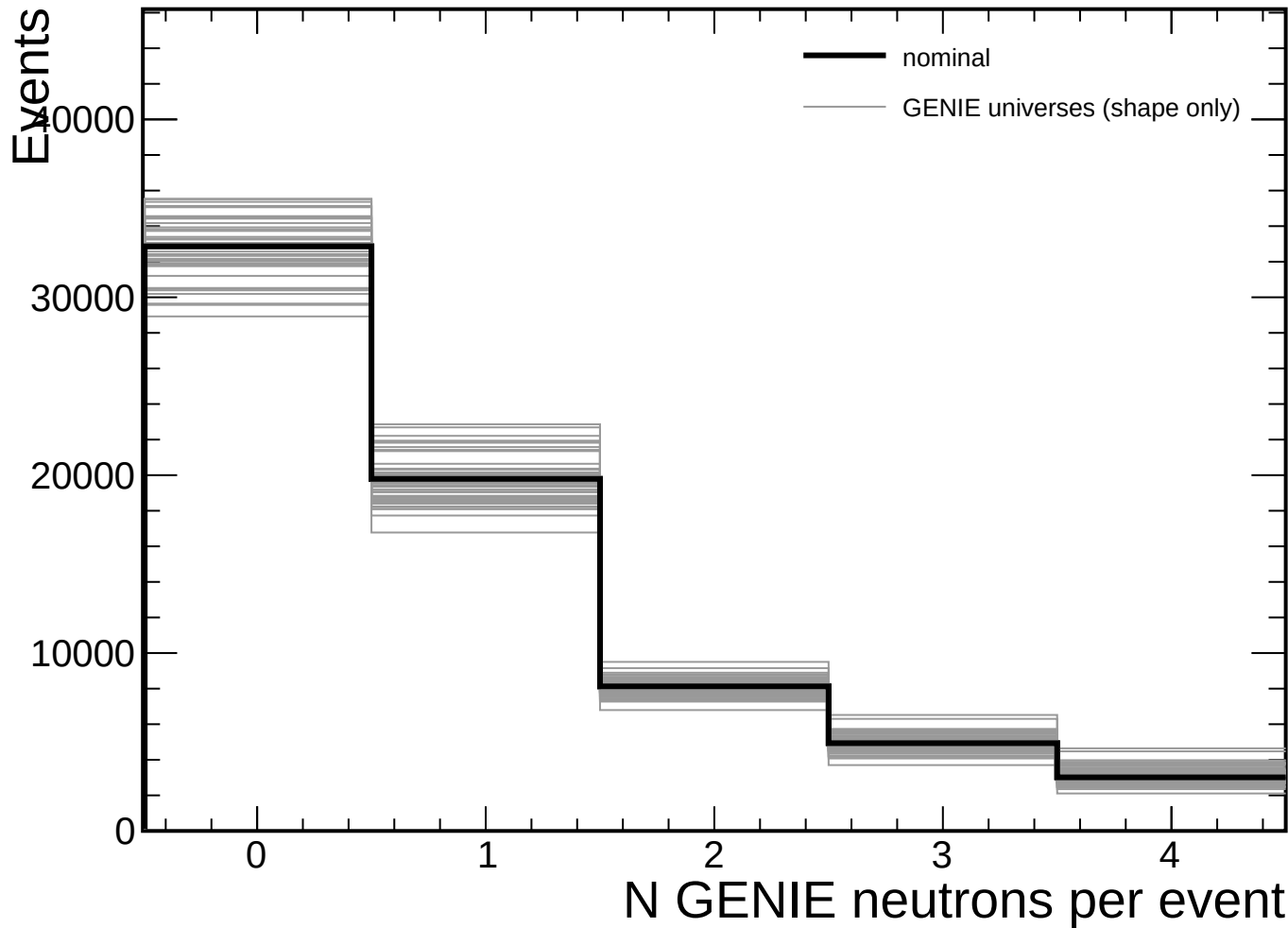


% diff from nominal

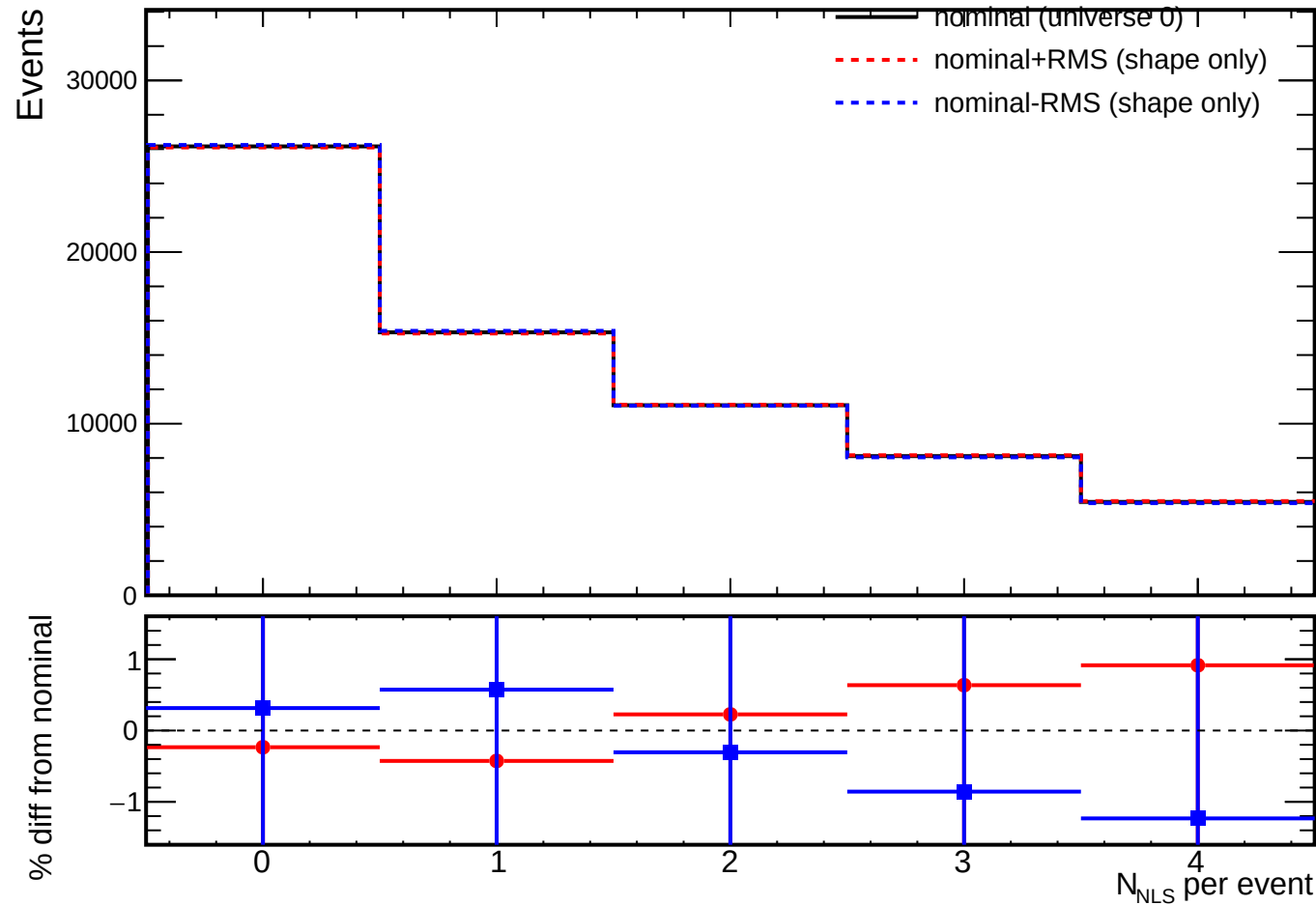


FHC: N GENIE neutrons GENIE multiverse (26 relevant systs, raw, 50 universes)

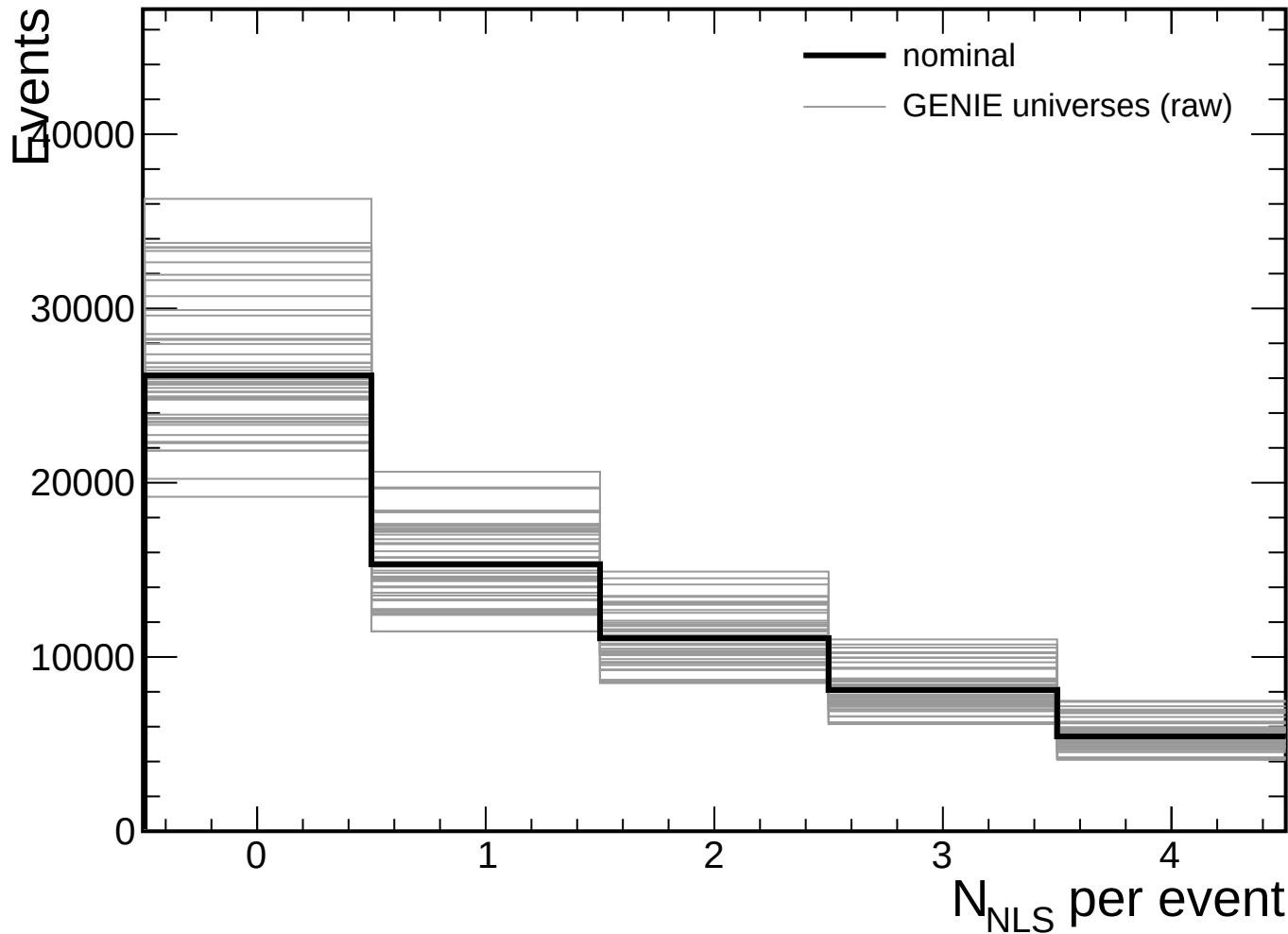


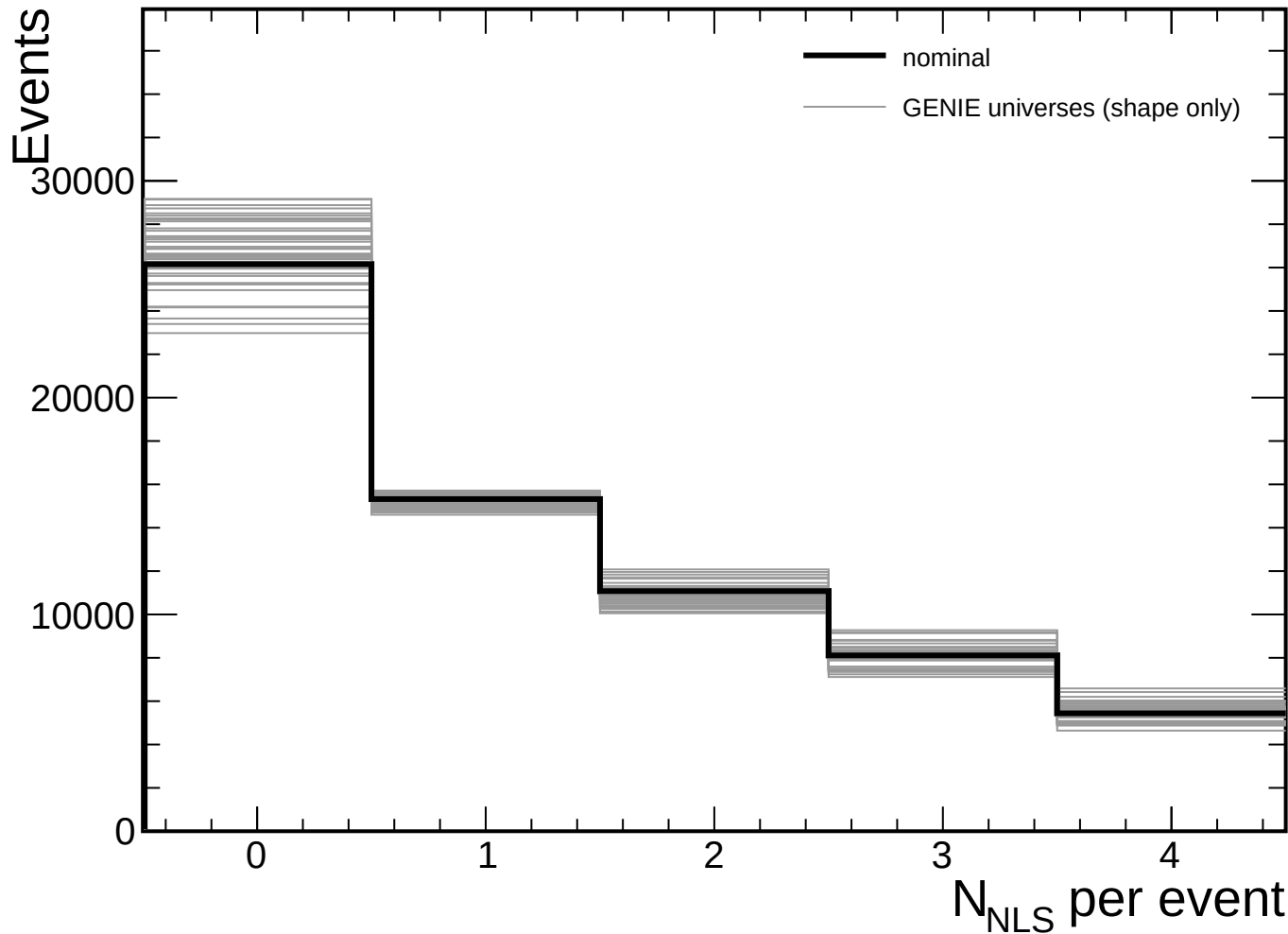


FHC: N_{NLS} per event -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=1.26$ $\chi^2_{\text{shape},-\text{RMS}}=2.29$ $\Delta\langle x \rangle_{+\text{RMS}}=+0.0051$ $\Delta\langle x \rangle_{-\text{RMS}}=-0.0069$
 $\chi^2_{\text{raw},+\text{RMS}}=1441.73$ $\chi^2_{\text{raw},-\text{RMS}}=1441.73$ rate: +14.76% / -14.76%



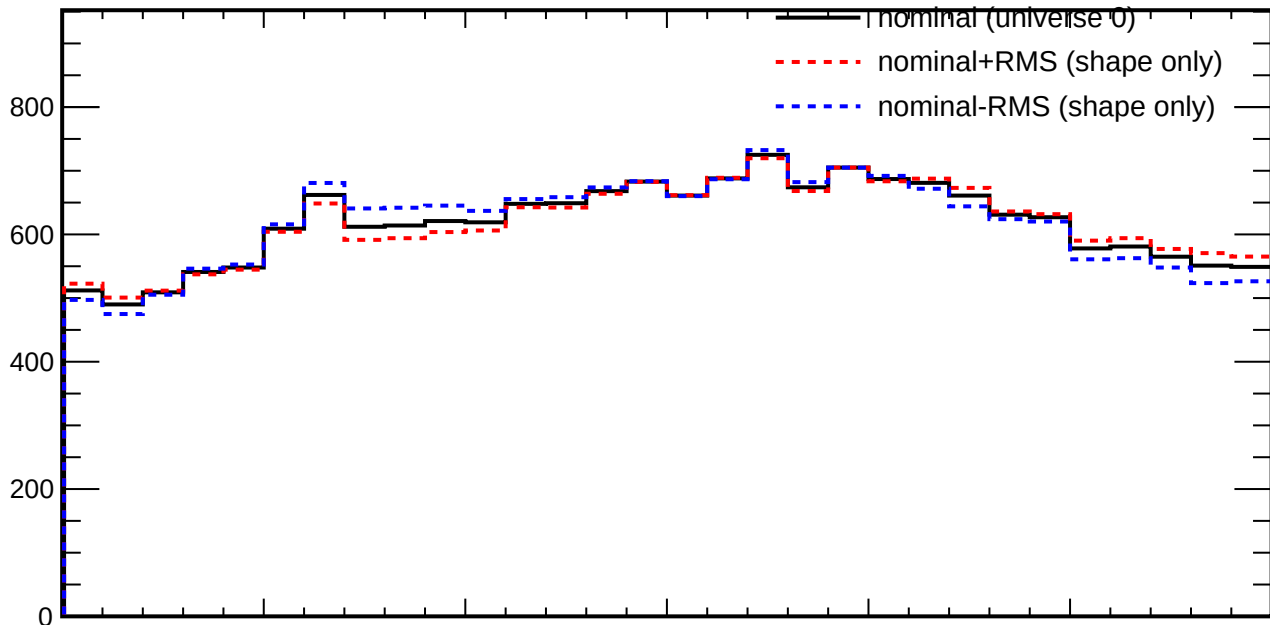
FHC: N_{NLS} per event GENIE multiverse (26 relevant systs, raw, 50 universes)



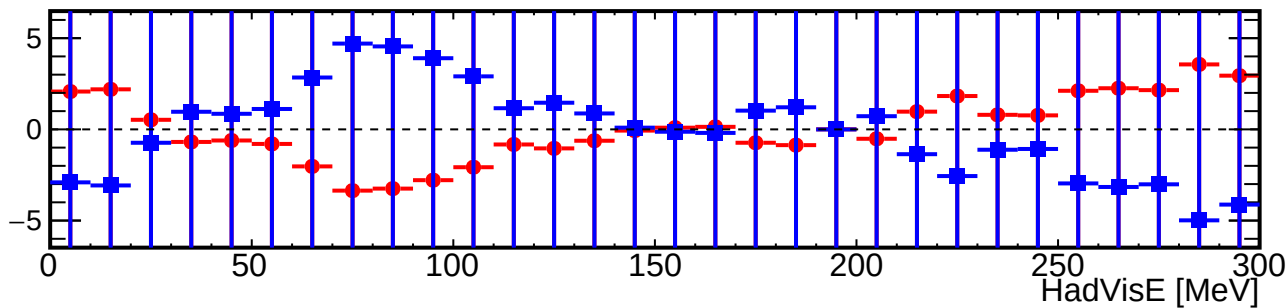


FHC: HadVisE [MeV] -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape,+RMS}}=5.52$ $\chi^2_{\text{shape,-RMS}}=10.82$ $\Delta\langle x \rangle_{\text{+RMS}}=+0.8010$ $\Delta\langle x \rangle_{\text{-RMS}}=-1.1219$
 $\chi^2_{\text{raw,+RMS}}=523.94$ $\chi^2_{\text{raw,-RMS}}=523.94$ rate: +16.69% / -16.69%

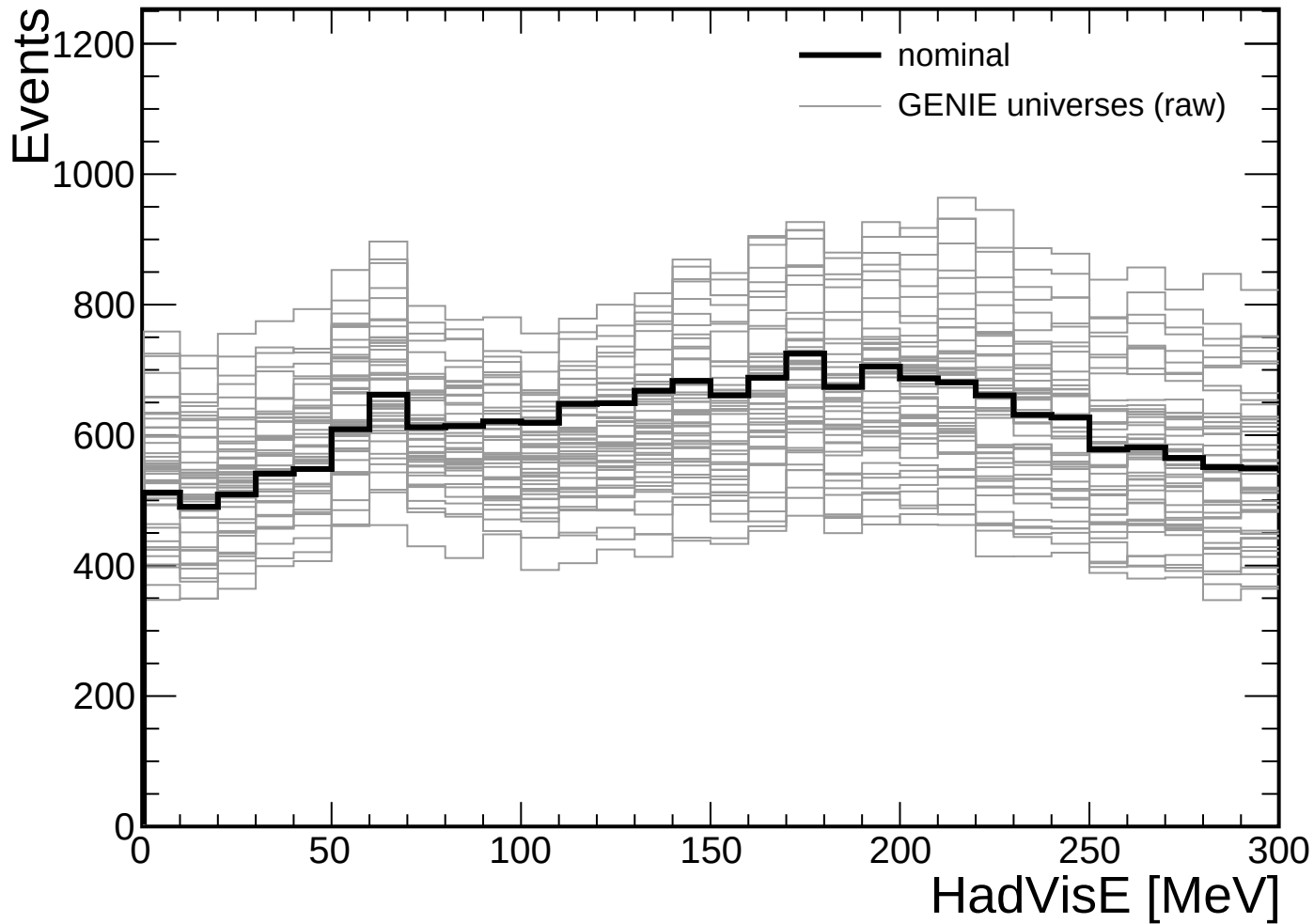
Events

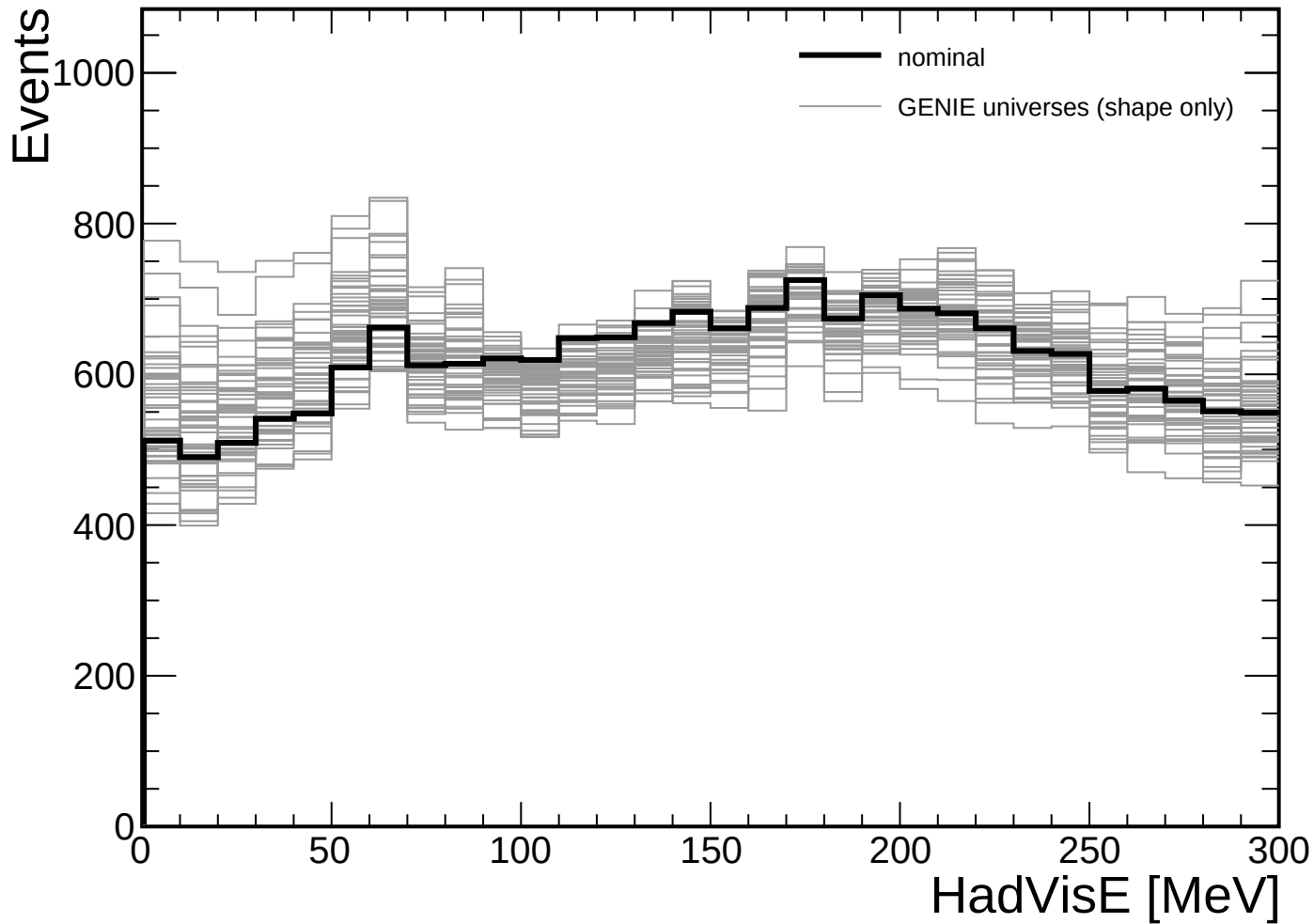


% diff from nominal

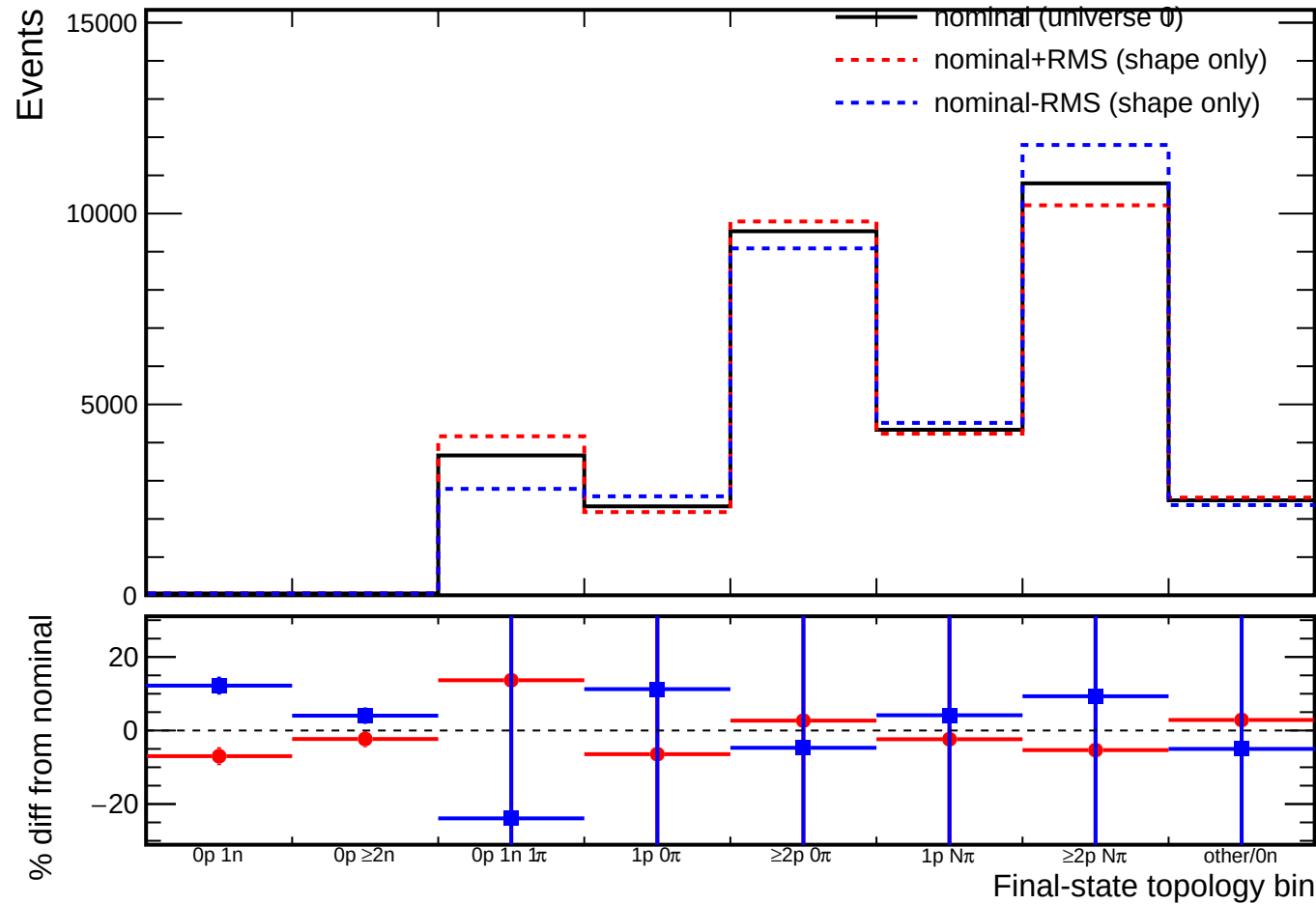


FHC: HadVisE [MeV] GENIE multiverse (26 relevant systs, raw, 50 universes)

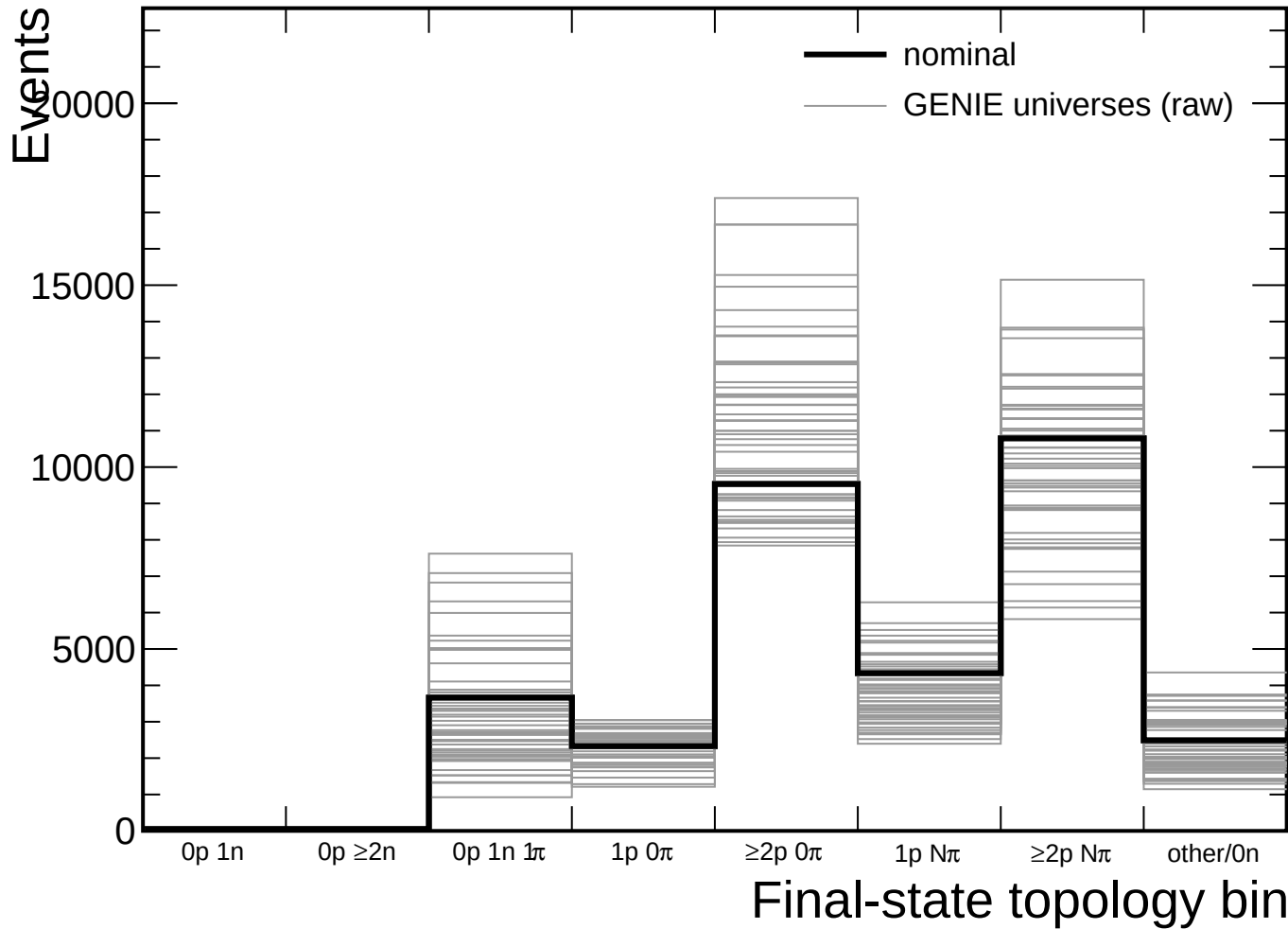


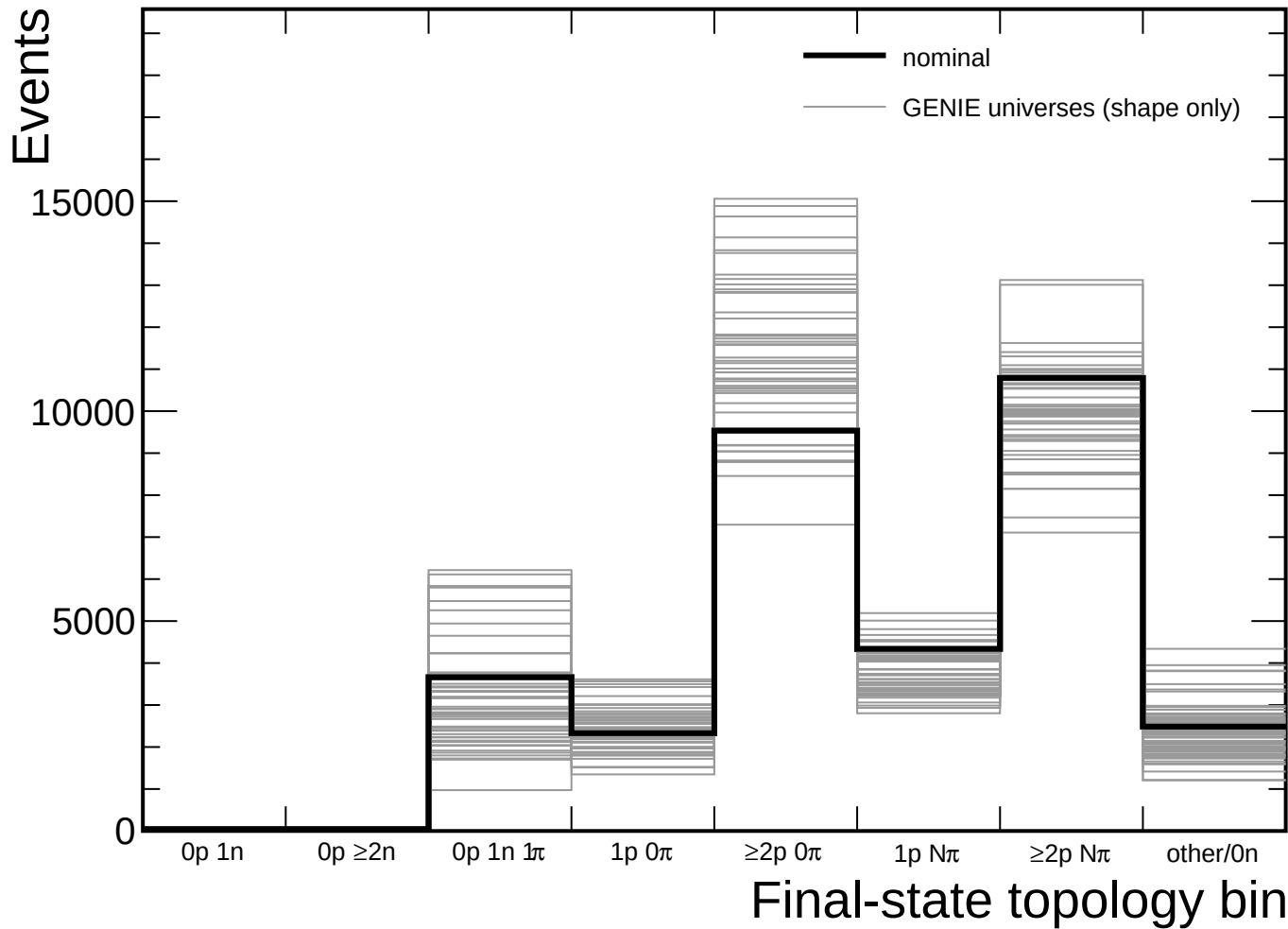


FHC: Final-state topology bin -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape}, +\text{RMS}} = 120.15$ $\chi^2_{\text{shape}, -\text{RMS}} = 367.47$ $\Delta(\chi^2)_{+\text{RMS}} = -0.0570$ $\Delta(\chi^2)_{-\text{RMS}} = +0.0996$
 $\chi^2_{\text{raw}, +\text{RMS}} = 2654.98$ $\chi^2_{\text{raw}, -\text{RMS}} = 2654.98$ rate: +27.24% / -27.24%

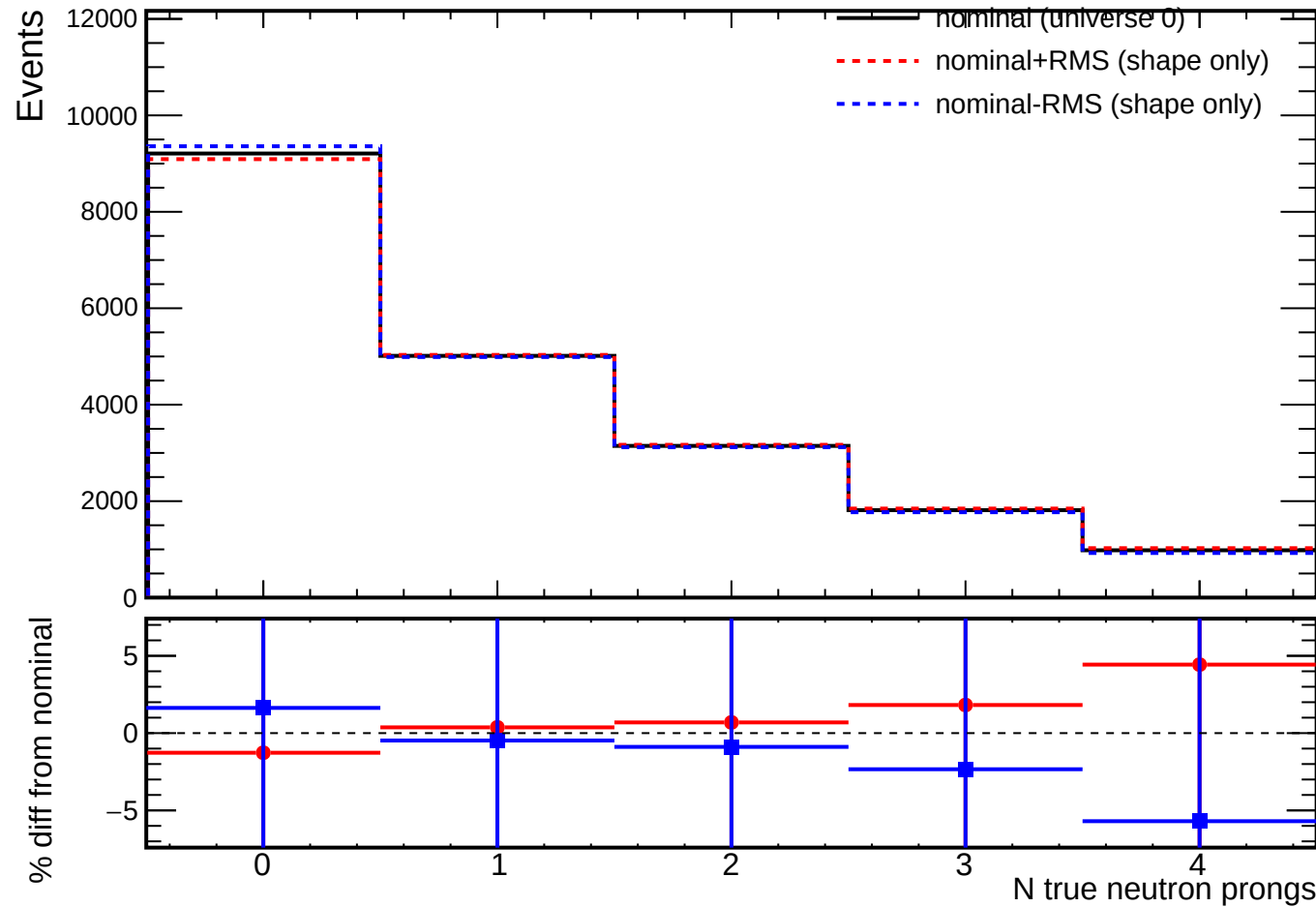


FHC: Final-state topology bin GENIE multiverse (26 relevant systs, raw, 50 universes)

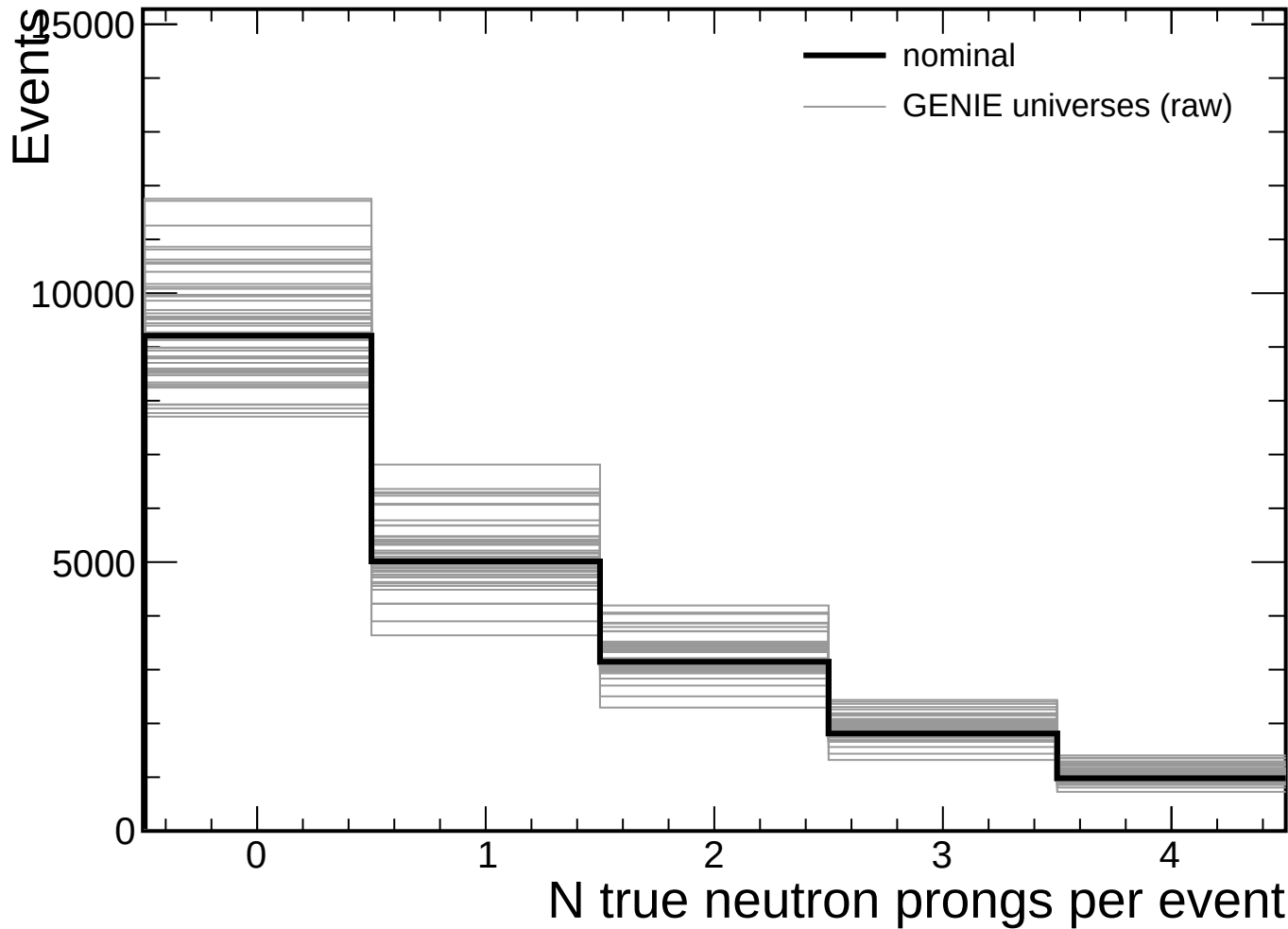


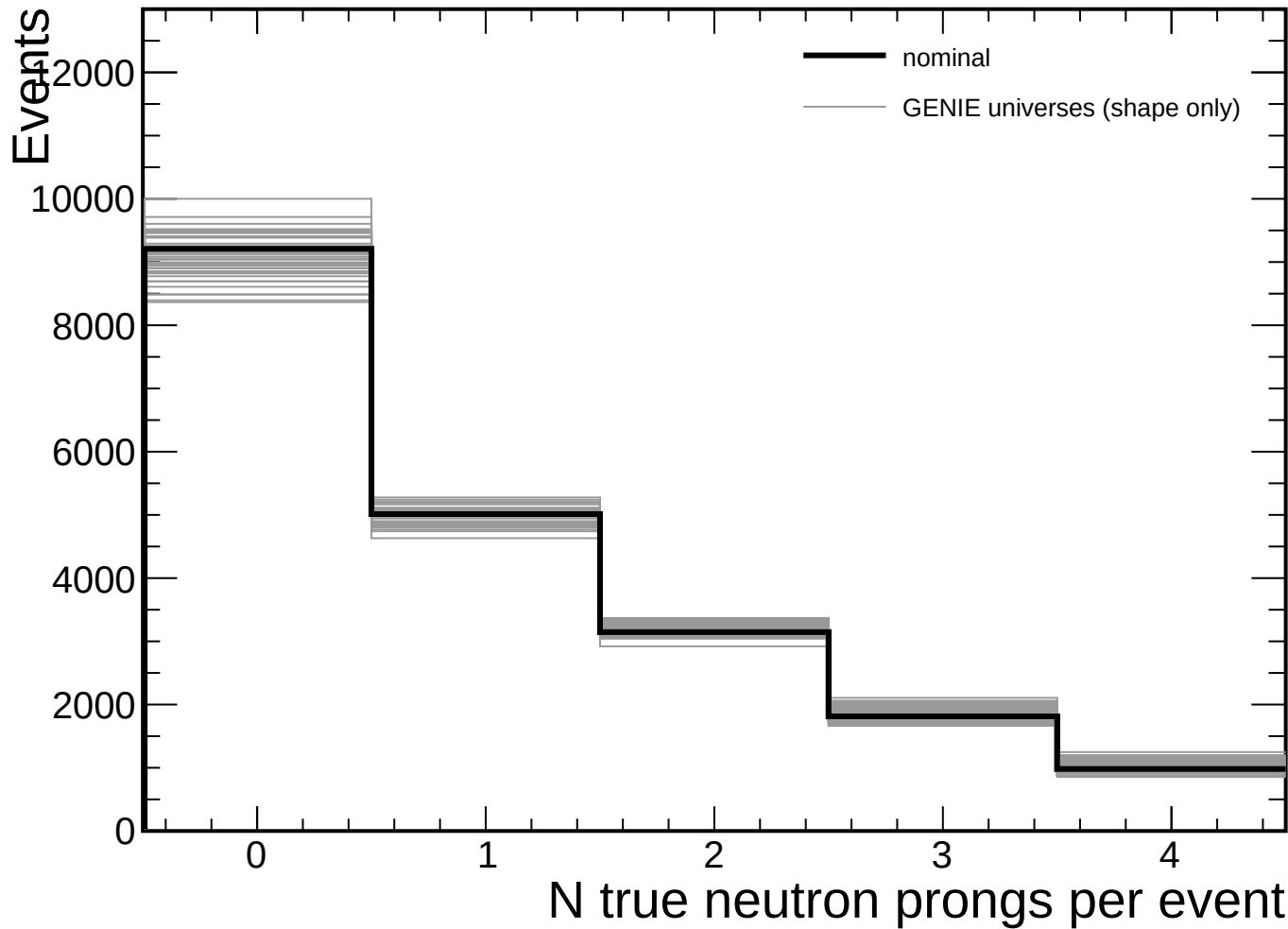


RHC: N true neutron prongs -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=4.23$ $\chi^2_{\text{shape},-\text{RMS}}=6.99$ $\Delta\langle x \rangle_{+\text{RMS}}=+0.0166$ $\Delta\langle x \rangle_{-\text{RMS}}=-0.0214$
 $\chi^2_{\text{raw},+\text{RMS}}=319.70$ $\chi^2_{\text{raw},-\text{RMS}}=319.70$ rate: +12.49% / -12.49%

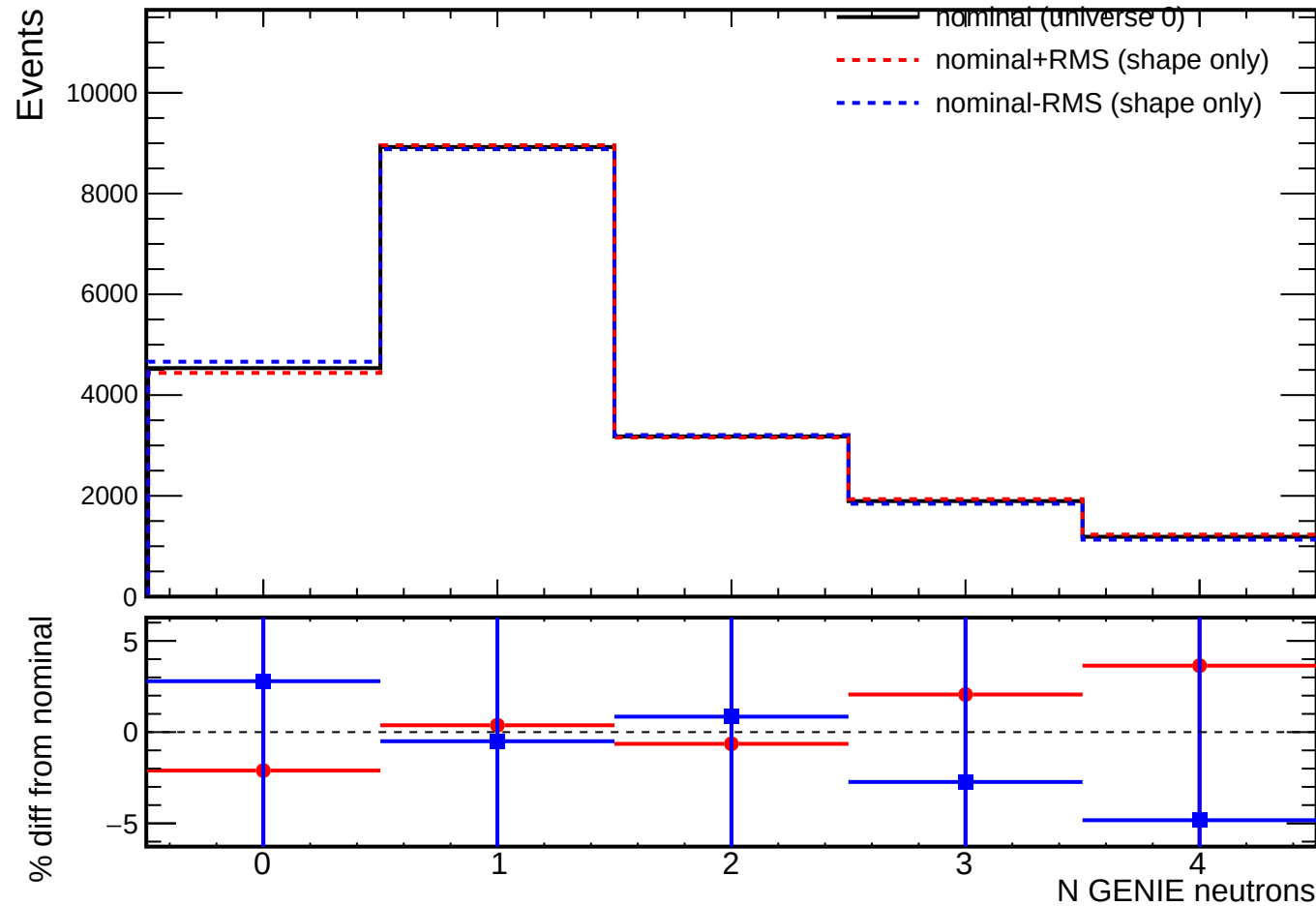


RHC: N true neutron prongs GENIE multiverse (26 relevant systs, raw, 50 universes)

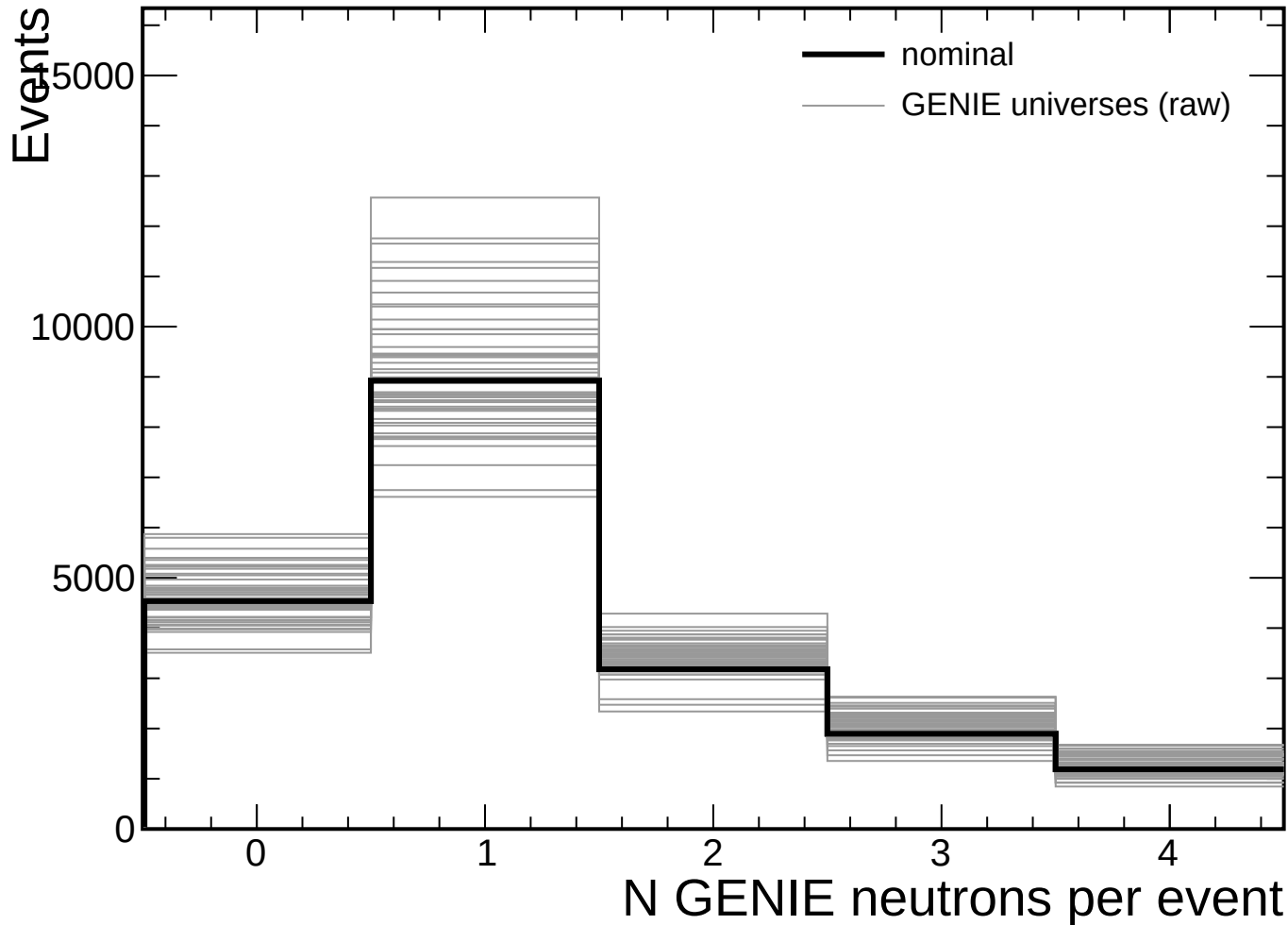


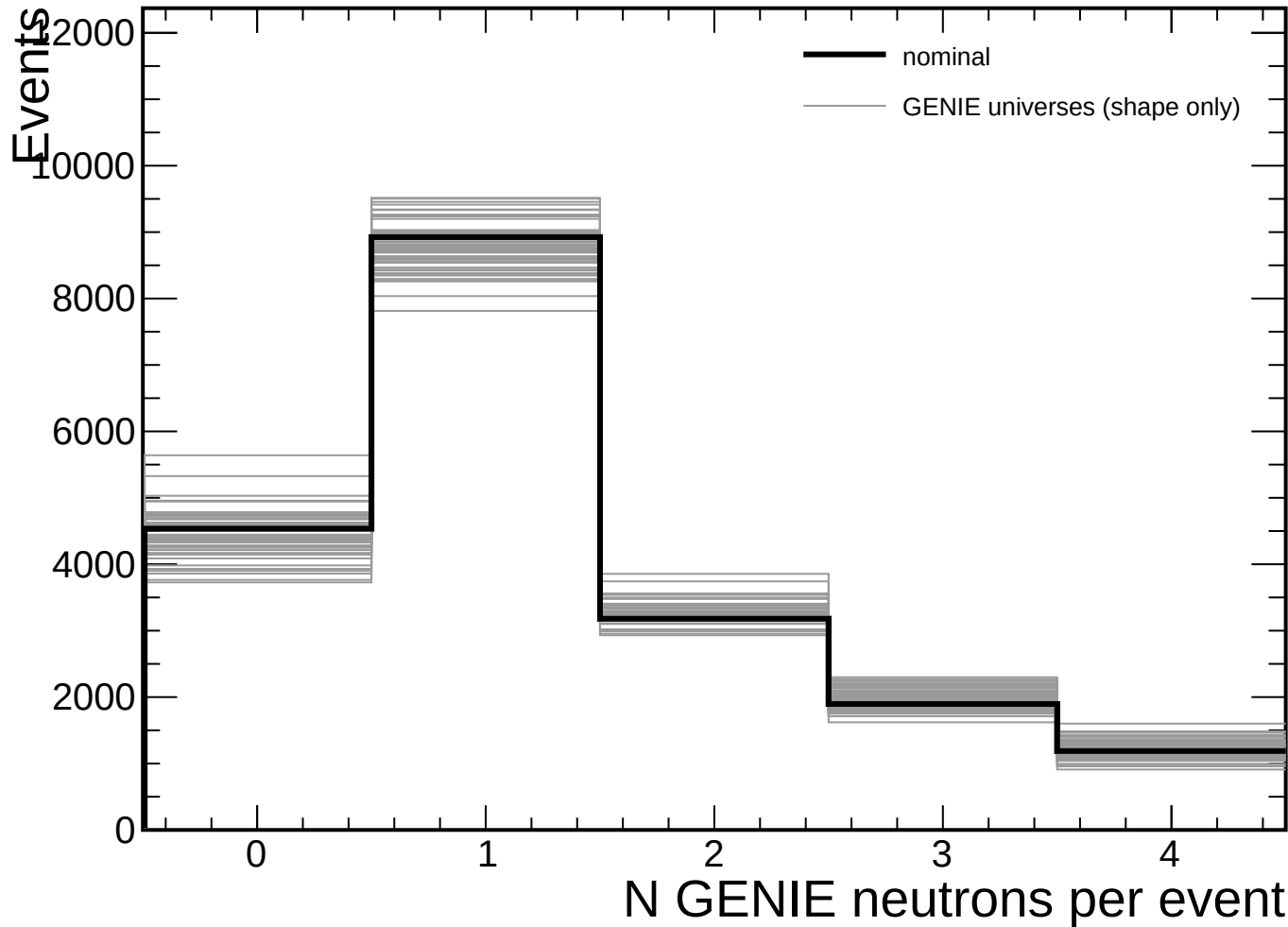


RHC: N GENIE neutrons -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=4.65$ $\chi^2_{\text{shape},-\text{RMS}}=8.17$ $\Delta\langle x \rangle_{+\text{RMS}}=+0.0144$ $\Delta\langle x \rangle_{-\text{RMS}}=-0.0190$
 $\chi^2_{\text{raw},+\text{RMS}}=391.44$ $\chi^2_{\text{raw},-\text{RMS}}=391.44$ rate: +13.98% / -13.98%



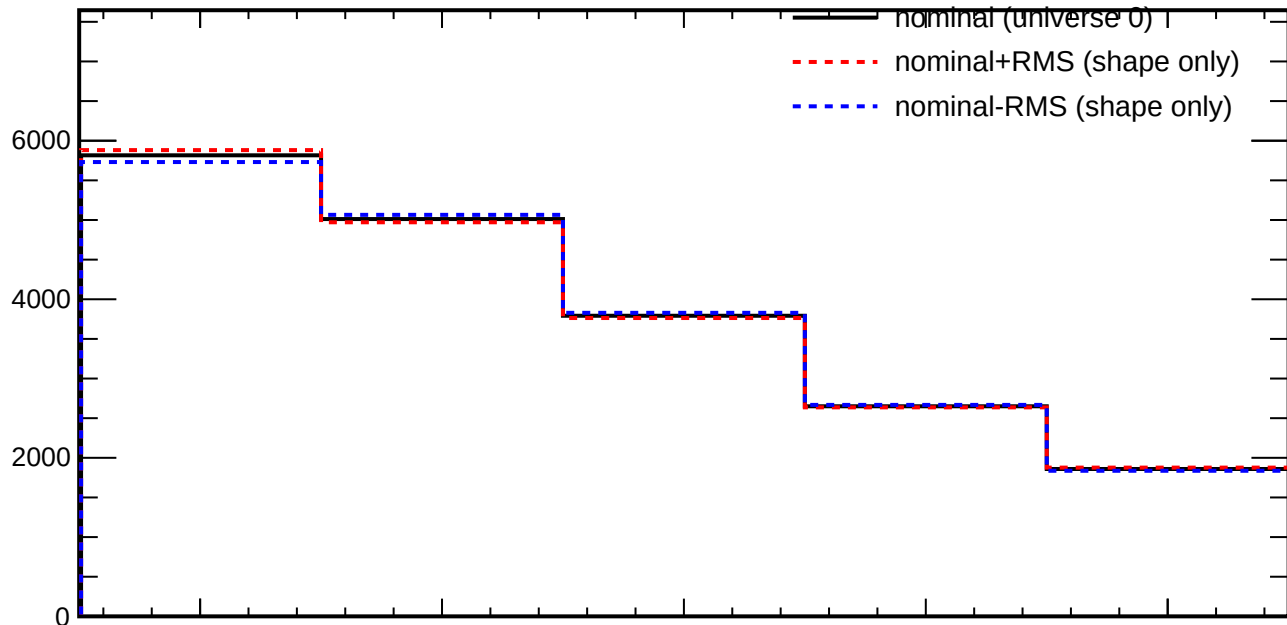
RHC: N GENIE neutrons GENIE multiverse (26 relevant systs, raw, 50 universes)



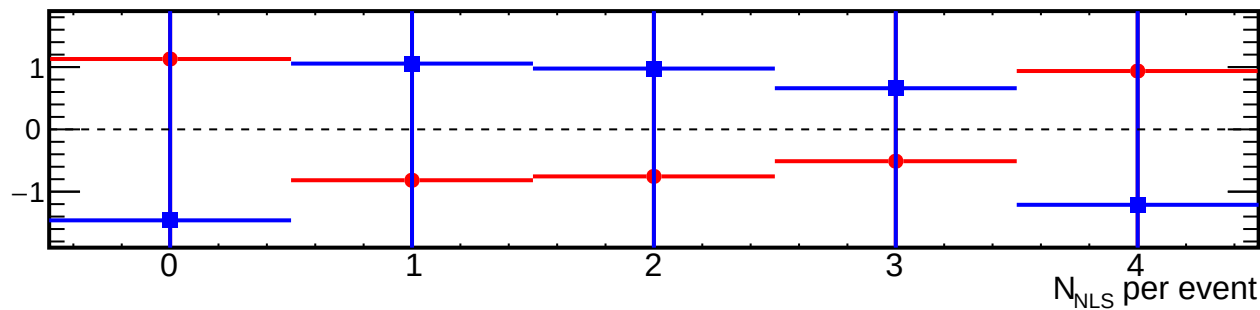


RHC: N_{NLS} per event -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=1.53$ $\chi^2_{\text{shape},-\text{RMS}}=2.55$ $\Delta\langle x \rangle_{+\text{RMS}}=-0.0036$ $\Delta\langle x \rangle_{-\text{RMS}}=+0.0047$
 $\chi^2_{\text{raw},+\text{RMS}}=314.17$ $\chi^2_{\text{raw},-\text{RMS}}=314.17$ rate: +12.78% / -12.78%

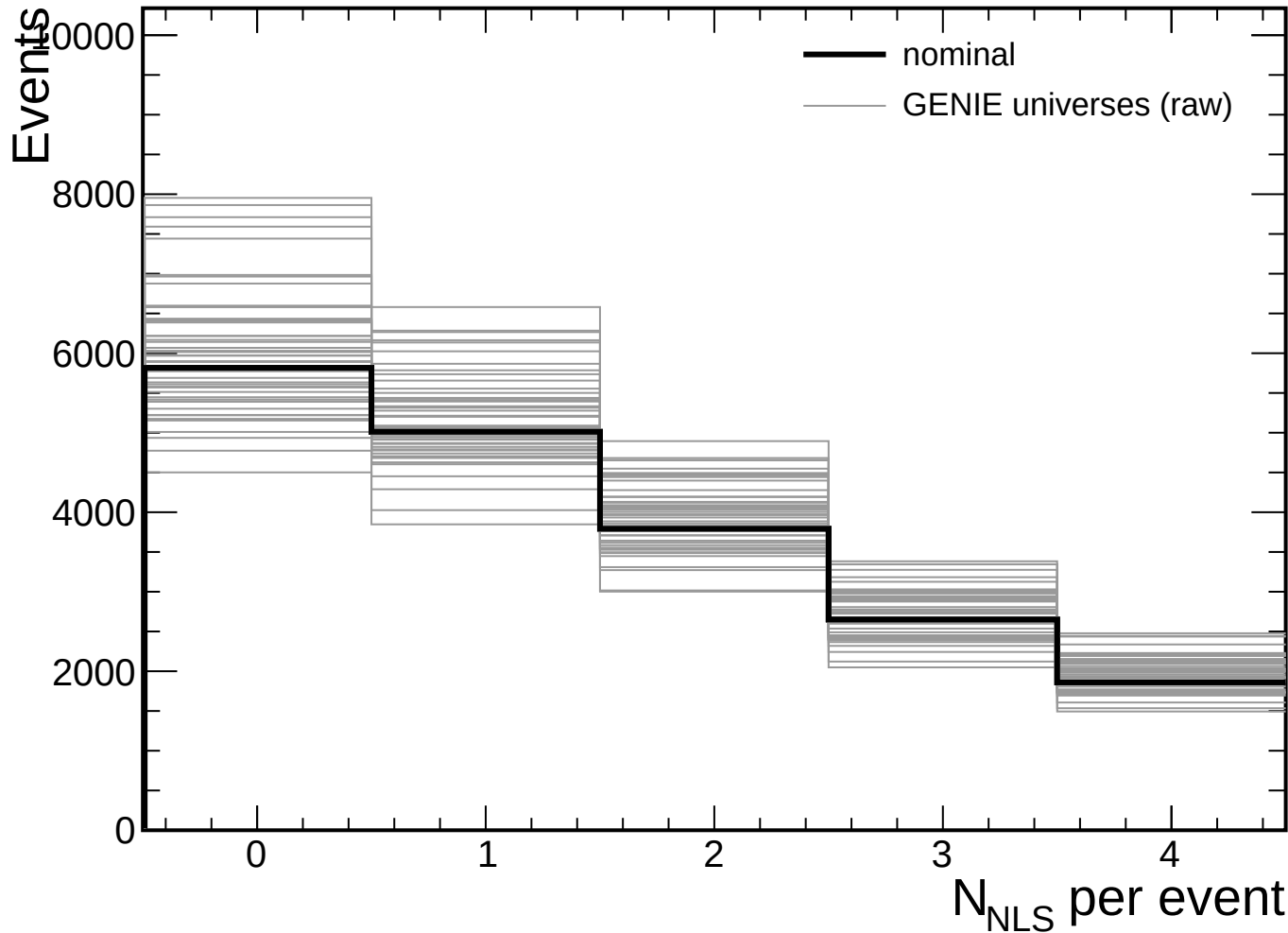
Events



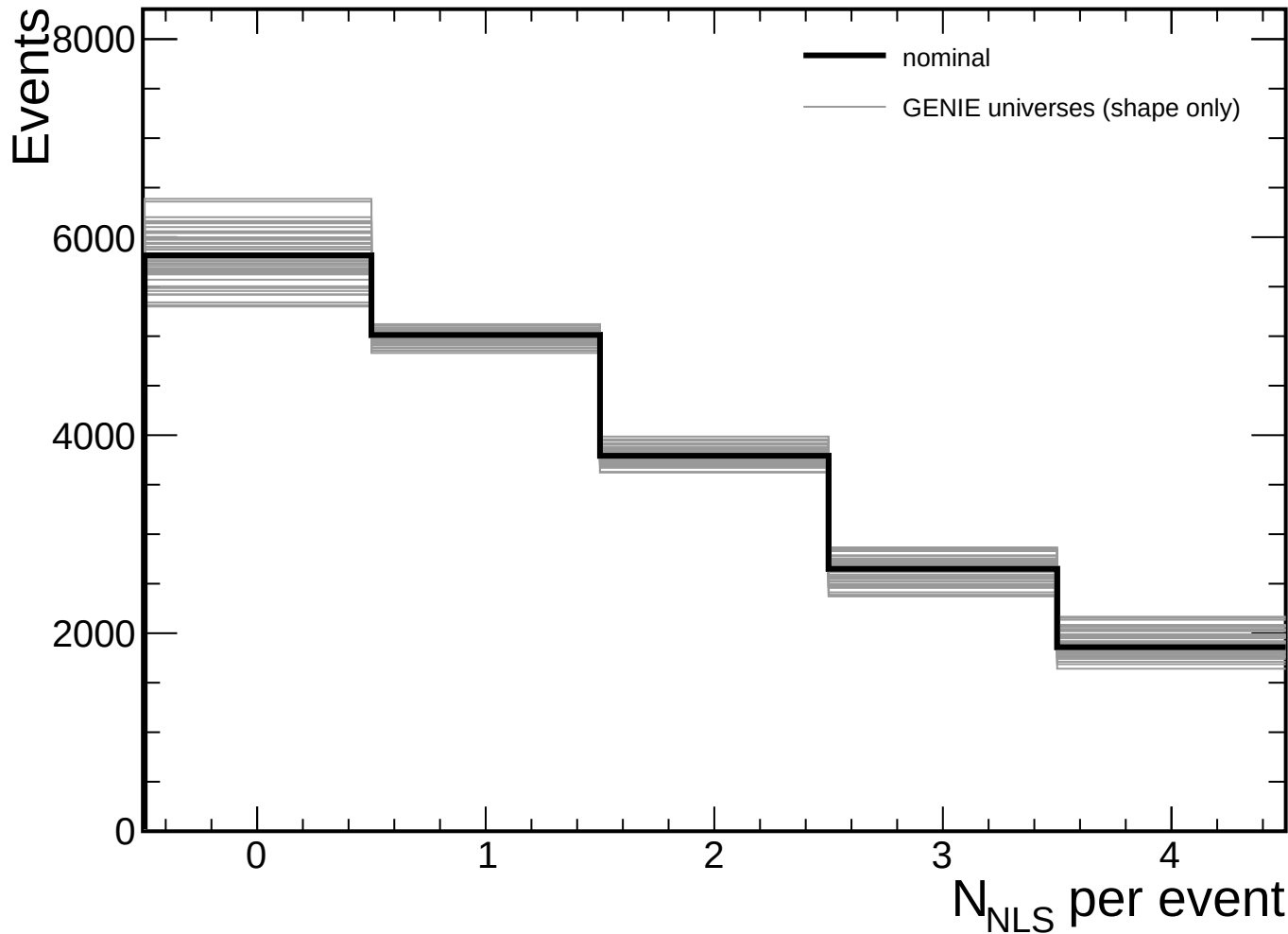
% diff from nominal



RHC: N_{NLS} per event GENIE multiverse (26 relevant systs, raw, 50 universes)

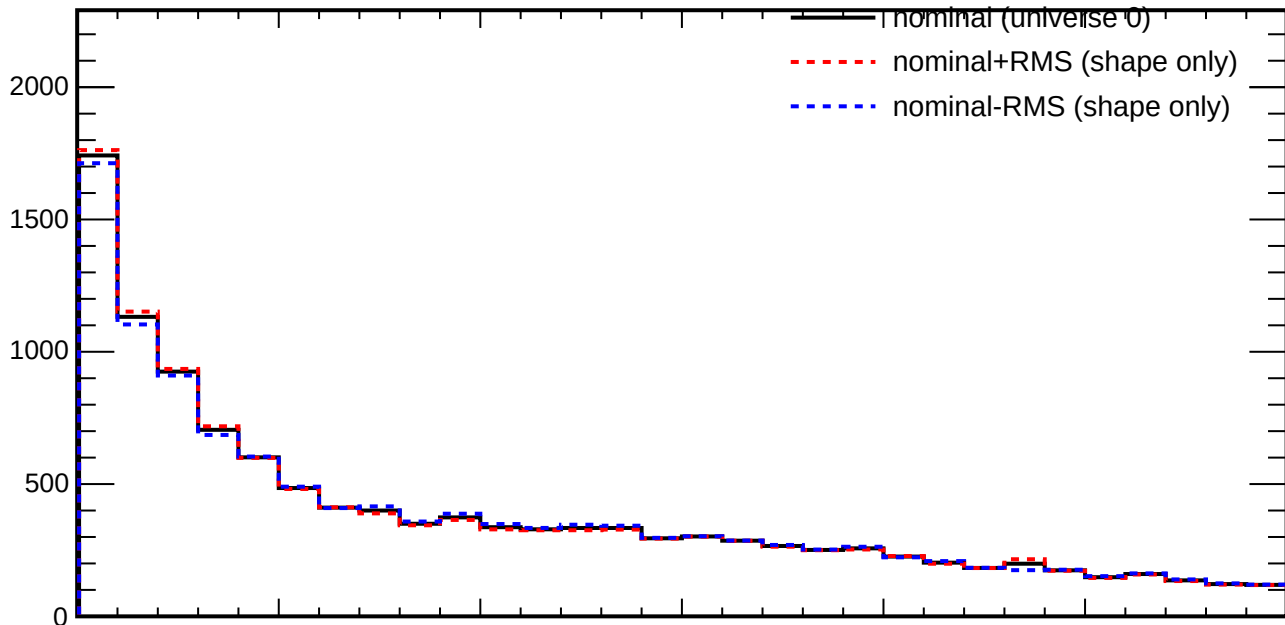


RHC: N_{NLS} per event GENIE multiverse (26 relevant systs, shape only -- each universe rescaled to nominal's total, 50 universes)

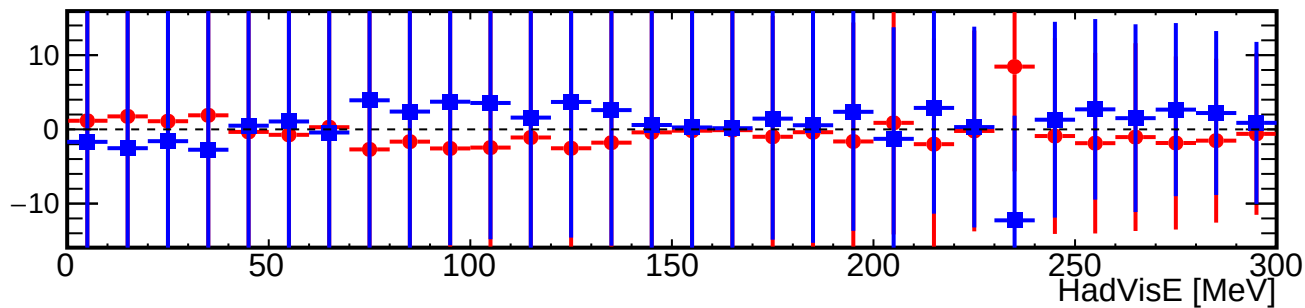


RHC: HadVisE [MeV] -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape}, +\text{RMS}} = 3.97$ $\chi^2_{\text{shape}, -\text{RMS}} = 8.34$ $\Delta\langle x \rangle_{+\text{RMS}} = -0.4903$ $\Delta\langle x \rangle_{-\text{RMS}} = +0.7107$
 $\chi^2_{\text{raw}, +\text{RMS}} = 402.69$ $\chi^2_{\text{raw}, -\text{RMS}} = 402.69$ rate: +18.36% / -18.36%

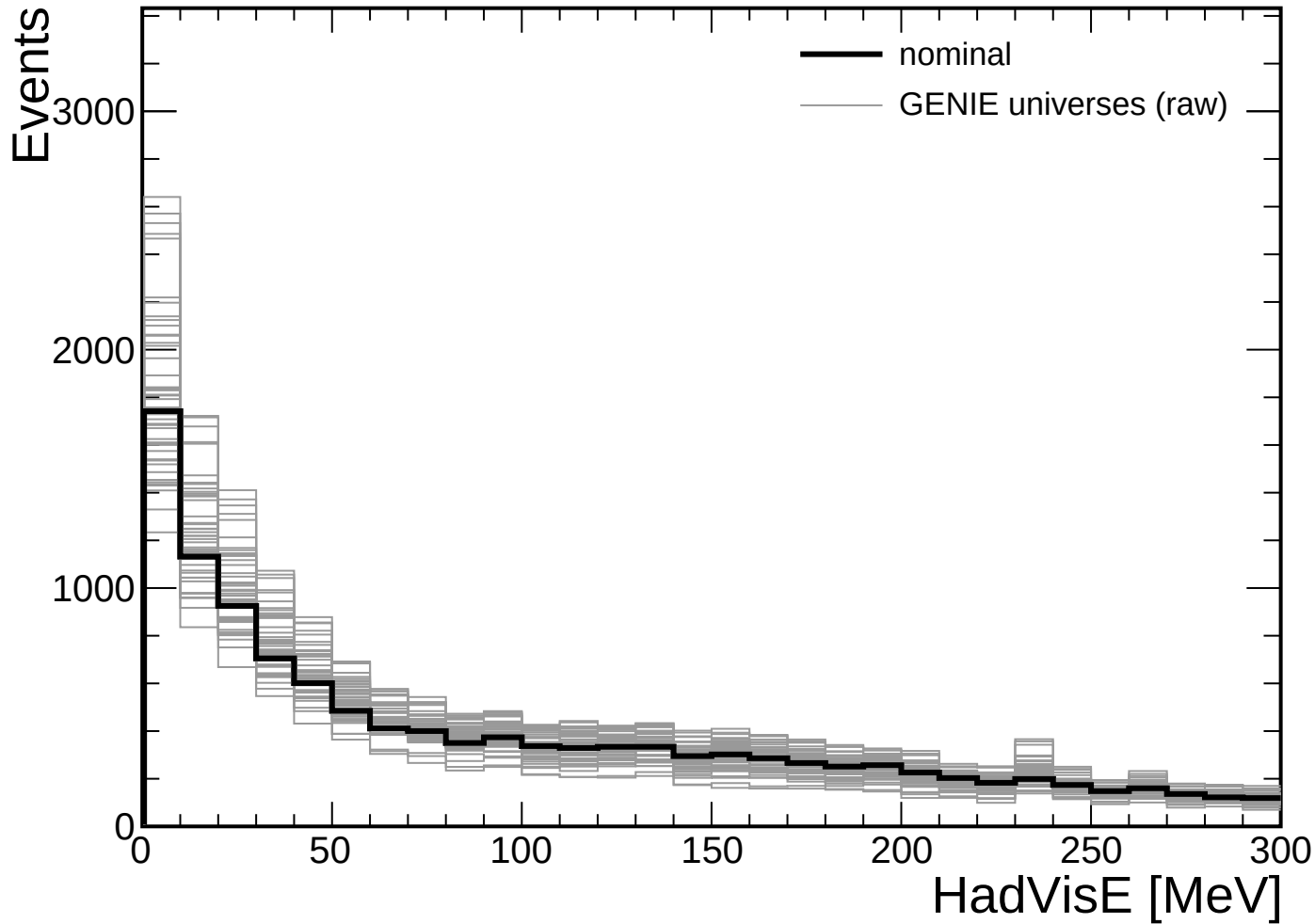
Events

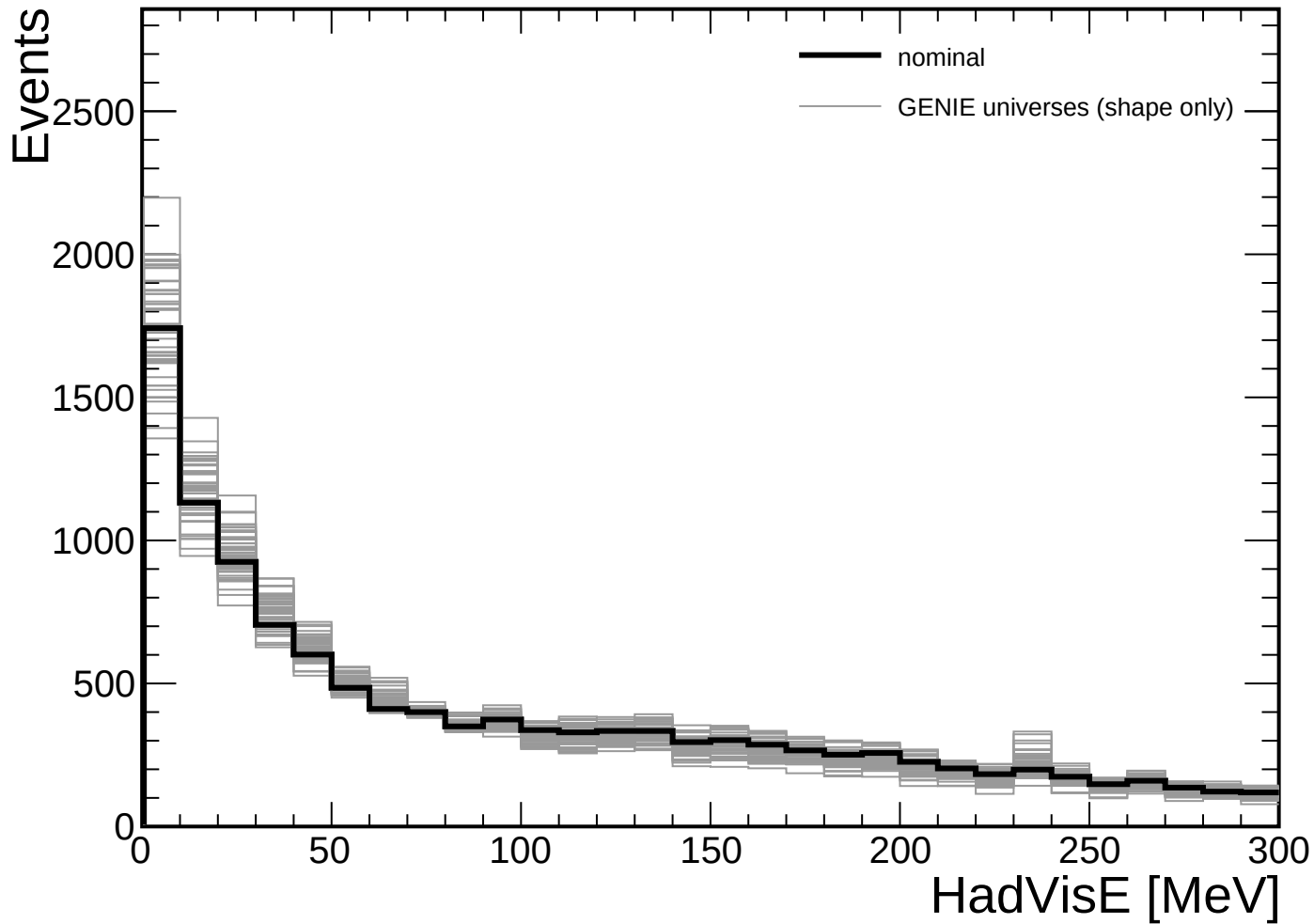


% diff from nominal

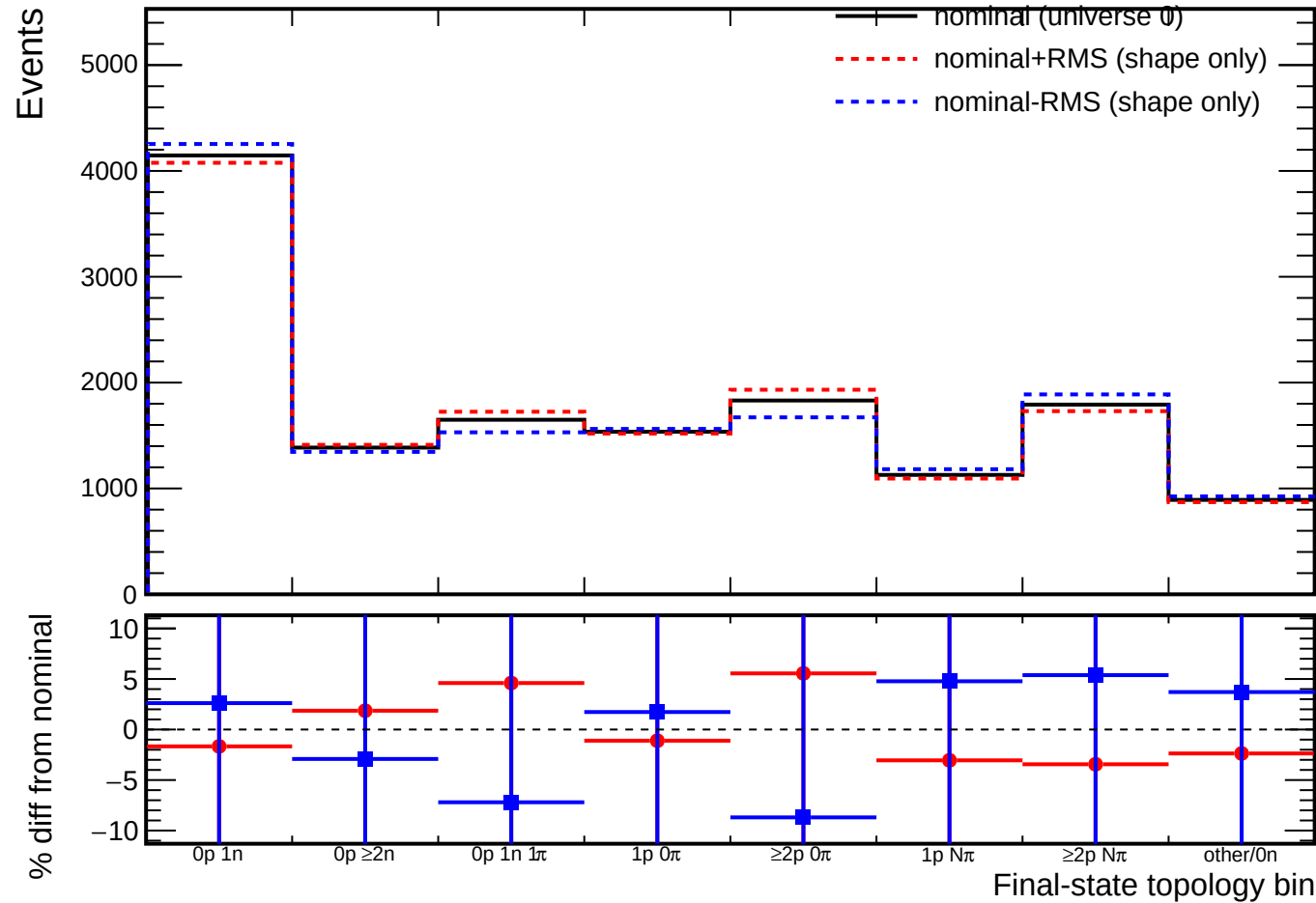


RHC: HadVisE [MeV] GENIE multiverse (26 relevant systs, raw, 50 universes)





RHC: Final-state topology bin -- GENIE multiverse (26 relevant systs, shape only) $\chi^2_{\text{shape},+\text{RMS}}=14.63$ $\chi^2_{\text{shape},-\text{RMS}}=35.85$ $\Delta\langle x \rangle_{+\text{RMS}}=-0.0109$ $\Delta\langle x \rangle_{-\text{RMS}}=+0.0170$
 $\chi^2_{\text{raw},+\text{RMS}}=719.05$ $\chi^2_{\text{raw},-\text{RMS}}=719.05$ rate: +22.04% / -22.04%



RHC: Final-state topology bin GENIE multiverse (26 relevant systs, raw, 50 universes)

